

Bocheng Cui

Curriculum Vitae

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RESEARCH INTEREST

Provenance and replication in distributed/cloud environments, as well as fault tolerance and edge systems.

PUBLICATION

Anthony Westbrook, **Bocheng Cui**, Lakshmi Ongolu, Elizabeth Varki. "Active Reproducibility for File-Based Exploratory Science", *Proceedings of the ACM Conference on Reproducibility and Replicability (ACM REP '26)*, 2026. [[PDF](#)]

Anthony Westbrook, Lakshmi Ongolu, **Bocheng Cui**, Elizabeth Varki. "Point-and-Click Reproducibility for Command-Line Data Analysis", *Proceedings of the ACM Conference on Reproducibility and Replicability (ACM REP '26)*, 2026. [[PDF](#)]

Bocheng Cui, Aleksey Charapko. "Simpler is Better: Revisiting Mencius State Machine Replication", *The 43rd International Symposium on Reliable Distributed Systems (SRDS)*, 2024. [[PDF](#)]

Owen Hilyard, **Bocheng Cui**, Marielle Webster, Abishek Bangalore Muralikrishna, Aleksey Charapko. "Cloudy Forecast: How Predictable is Communication Latency in the Cloud?", *2024 33rd International Conference on Computer Communications and Networks (ICCCN)*, 2024. [[PDF](#)]

EXPERIENCE

Jun 2026 — Aug 2026

Title: Software Development Engineer

Company: Evolv

- Developed a pre-cut algorithm for virtual 3D imagery, resolving errors in projection onto 2D camera images.
- Designed and implemented a fault-tolerant camera processing pipeline, with a modest improvement in frame rate (FPS).

Aug 2021 — Present

Title: Teaching Assistant

Institution: University of New Hampshire

Department: Computer Science

Served as a teaching assistant for both graduate and undergraduate courses.

Jun 2024 — Aug 2024

Title: Research Associate

Institution: University of New Hampshire

Department: NOAA-UNH Joint Hydrographic Center

Develop a converter from Tiny 1.2 to IHO SVG Schema and Translate python scripts from 2.7 to 3.6.

May 2022 — Aug 2022

Title: Research Assistant

Institution: University of New Hampshire

Department: Computer Science

Conducted evaluations and optimizations of existing consistency protocols, focusing on improving efficiency and reliability in distributed systems.

May 2022 — Aug 2022

Title: Cloud Software Developer

Institution: University of New Hampshire

Department: NOAA-UNH Joint Hydrographic Center

Implementation and testing of cloud based software and architectures for hydrographic and exploration bathymetric mapping.

SKILLS

Languages: Go, C/C++, Python, Java, SQL, R, Bash.

Systems & Infrastructure: Docker, Kubernetes, OpenWhisk, Mininet, gRPC, AWS/GCP/Azure, Linux, HPC clusters, Slurm, Git, Conda, LaTeX.

AI-Assisted Development: Claude Code, GitHub Copilot; LLM API integration (Anthropic, OpenAI); prompt design for code generation, review, and testing.

Domains: Distributed systems, cloud and serverless computing, computer networks, file systems.

PROJECTS

Provenance file system in distributed platform

- Designing and developing a provenance file system for cloud platforms, aimed at providing enhanced tracking and management functionalities for cloud environments.

Mencius Revisited

- Conducted an in-depth study of state machine replication protocols, including Mencius, E-Paxos, and Paxos, with a focus on enhancing the Mencius protocol.
- Implemented optimizations in the failure handling mechanisms of the Mencius protocol, significantly increased max throughput by improving the protocol.

Cloud latency test

- Designed and implemented a comprehensive tool for testing latency in cloud environments.
- Conducted thorough evaluations and studies on latency performance and stability across various Cloud Providers, including AWS, GCP, and Azure.

Fault Tolerance Sharded K/V Storage System with MapReduce

- Engineered a fault-tolerant K/V storage system, implementing Raft for reliability, and a sharding algorithm with customizable shard placement for efficient large-scale data handling.
- Developed a non-disruptive data migration algorithm and integrated Google MapReduce for enhanced data processing and analysis.

EDUCATION

Ph.D. in Computer Science, <i>University of New Hampshire</i>	Since 2022
Master. in Computer Science, <i>University of New Hampshire</i>	2018 - 2022
Bachelor of Material Science, <i>Qingdao University of Science & Technology</i>	2014 - 2018

ACHIEVEMENTS

UNH Summer TA Fellowship	2025
UNH College of Engineering and Physical Sciences (CEPS) Dean Scholarship	2018 - 2022
Vice President of the Students Union at Qingdao University of Science and Technology	2015 - 2017